AI Foundations & Modern AI Landscape

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1.Introduction to Artificial Intelligence

Definition

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence, such as learning, reasoning, problem-solving, and decision-making.

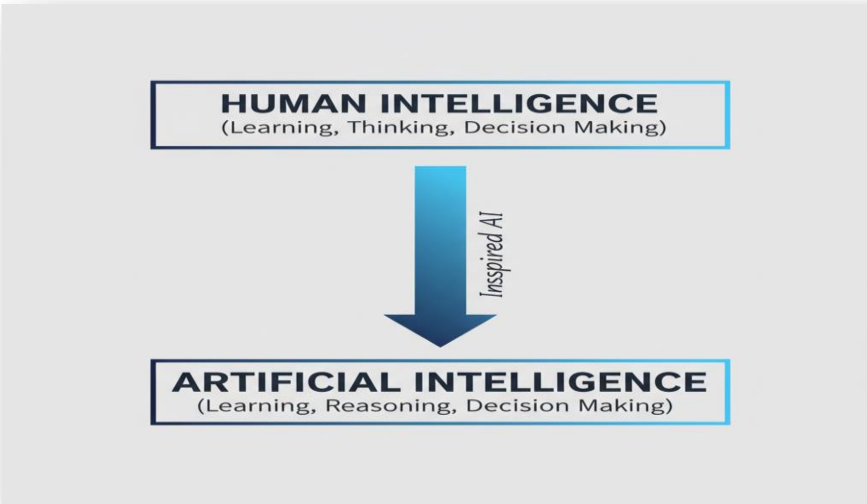
->In simple words we can say think of it as a **computer that can learn from experience, just like a human.**

Real-Life Examples

- ChatGPT answers questions.
- Google Maps finds the best route.
- YouTube recommends videos.
- Face Unlock recognizes faces.
- Alexa responds to voice commands.

Why is AI Important?

- Automates repetitive tasks.
- Saves time and effort.
- Improves accuracy.
- Helps in decision-making.
- Used in healthcare, education, finance, and transportation.



2. AI vs Human Intelligence

->However, unlike humans, **AI does not possess emotions, consciousness, or true understanding.**

When you feel:

- Happy after winning a prize
- Sad after losing a friend

->These feelings come from your brain, hormones, personal experience, when ChatGPT says (I'm sorry you are feeling bad). It has been learned that when someone says they are sad, this is an appropriate response. So, AI can **recognize emotions** and **talk about emotions**, but it does not **experience emotions**.

What is Consciousness?

->Being aware of yourself and your surroundings. (Like Your name, your memories, your goals, that you exist)

->Right now, you are aware that you are reading this message. That awareness is called consciousness.

->AI does not have this. AI does not know:

- Who it is
- Why it exists
- Whether it exists

->It simply processes inputs and generates outputs.

Why AI Does Not Have True Understanding

Let's see an example.

You read:

"The boy dropped the glass because it was slippery."

You understand:

- What a boy is
- What a glass is
- What slippery means
- Why the glass fell

->You connect the sentence with real-world experience.

->AI does something different; **it learns patterns from huge amounts of text.**

It predicts:

"After these words, these words usually come next."

->**AI often appears to understand. But underneath, it is mostly finding patterns in data**.

3. AI Around Us

->Many applications and services use AI to improve user experience and provide intelligent solutions.

Examples are --->ChatGPT, YouTube, Netflix, Google Maps, Self-Driving Cars etc....

->AI uses **data, algorithms, machine learning models and computing power**. **Without algorithm AI cannot work**

-> for example, if we take YouTube:

Data collections:

->Data is collected in a way of watch history, search history and time spent on which content, likes, subscription, comments. This is how the data is collected on YouTube.

Algorithms:

-> Algorithms are simple set of instructions

For example --> if user is watching more AI video, then recommend more AI videos

-> YouTube has billions of users, manually writing script for individual persons are impossible

So here was machine learning to do its work.

Machine learning:

->Machine learning is used to see the patterns

if user A watches AI, he also watches python for it

If user B watches AI, he also watches python for it

If user C watches AI, he also watches python for it

-> The ML learns that people who watch AI also watch python. **No one programmed that rule for it. It learned that**. This is called machine learning.

Model Training:

-> Training means learning.

->Suppose YouTube has1 billion users,100 billion videos watched

->The ML model studies all this data.

It asks:

- What videos do people watch?
- What do they like?
- What do they click on?

->The model adjusts itself until it becomes good at prediction. This process is called Training

Inference:

-> inference is the process where a trained model uses its learned patterns to make **predictions** or decisions on new, unseen data for example-> This user likes AI. Show this video.

Computing power:

-> computer's ability to process massive amounts of information and perform complex mathematical calculations extremely fast.

For example--> If you have a massive library but read at the speed of a snail, it will take centuries to learn anything. With high computing power, an AI can "read" that entire library in a few days, allowing it to learn languages, create art, or drive cars.

4. EVOLUTION OF AI:

Before AI existed (1940-1950):

->can machines think like humans, at that time computers were very basic. One of the scientists called **Alan Turing** conducted **Turing test**

->**TURING TEST: a human talks to a human and a machine If the human cannot tell which is which, the machine may be considered intelligent.**

->This idea became one of the foundations of AI.

2. Birth of AI (1956):

->The term **Artificial Intelligence** was officially introduced in 1956 by:John McCarthy

At the famous: Dartmouth Summer Research Project on Artificial Intelligence

->Scientists believed machines would soon: Solve problems, understand language, Learn like humans

->This event is considered the official birth of AI.

3. AI Winter (1960–1970):

->technology and data availability limitations reduced funding and interest in ai research

->AI did not make big headlines for decades until 1977

4. Early AI Success (1956–1974):

->Researchers created programs that could: Play Games, Programs learned to play

- Checkers
- Chess

->Solve Mathematical Problems: Programs could solve logical puzzles.

Example: May 1997, when IBM's supercomputer **Deep Blue** defeated Grandmaster **Garry Kasparov** in a historic six-game match in New York City. This victory marked the first time a machine beat a human world champion under standard tournament time controls.

5.Advancement of Technology (1990–2000):

->computer significantly increased in power and the internet rapidly expanded across the globe

->People worldwide were churning out massive volumes of digital data and computer processing capabilities skyrocketed

->this dramatic shift proved to be a turning point for AI Research

6.Deep learning (2006):

->**Geoffrey Hinton** co-published a paper in **2006** reviving the interest in **neural networks with deep learning techniques**

->**Deep learning neural networks are a type of AI inspired by how the human brain works**

->One needs lots of data and high computational power to run deep learning models

7.IBM's Watson(2011):

->**IBM's Watson AI won on the jeopardy quiz show** showcasing substantial advancements in natural language processing

->AI was able to work with human language

8.Deep Neural Networks (2012):

->researchers at Stanford and google including Jeff Dean and Andrew Ng – published building high level features using large scaled unsupervised learning

->A paper that uses multilayer neural nets. Their system excelled at recognizing cats

9.Google Brain (2017):

->The Google Brain team developed a fundamental scientific contribution introducing transformers

-> this model was significant breakthrough in natural language processing due to its use of self-attention mechanism allowing it to process data sequences like text efficiently

10.Generative AI Era (2020–Present):

->AI moved from Recognizing o Creating Large Language Models (LLMs) became popular.

Examples:

- ChatGPT

- Gemini
- Claude

These systems can:

- Generate text
- Write code
- Create images
- Summarize documents
- Answer questions

This is the era we are currently living in.

5. Machine Learning VS Deep Learning:

Machine Learning:

Machine Learning is a method that allows computers to learn patterns from data and make predictions without being explicitly programmed.

Example: Predicting Student Results

Suppose you have data:

Study Hours	Result
1	Fail
2	Fail
5	Pass
8	Pass

So, machine leaning learns that:

More study hours
↓
Higher chance of passing

->Now if a new student studies 7 hours =ML predicts: Pass

->Nobody manually wrote:

->IF study hours > 4, THEN Pass, the machine learned the pattern itself.

Deep Learning:

->Deep Learning is a special type of Machine Learning that uses **Neural Networks** with many layers.

->It is inspired by how neurons work in the human brain.

Example: Face Recognition

->Suppose you want AI to recognize your face, A normal ML model struggles because faces are very complex.

->Deep Learning automatically learns:

Eyes
↓
Nose
↓
Mouth
↓
Face Shape
↓
Complete Face

->Distance between eyes, Face shape, Nose structure, Jaw line then it predicts match found unlock phone

Why was deep Learning found?

->Machine learning works well for tables, numbers, small data sets but it struggles with videos, images, speech, language

->It was created to solve complex problems

Easy Analogy

Imagine teaching a child to identify cats.

Machine Learning

You tell the child: Cats have, Whiskers, Tail, Pointed ears

The child follows those features.

Deep Learning

You show:

100,000 Cat Photos The child automatically discovers: Whiskers, Ears, Eyes, Tail, Shape

->without you telling anything.This is how Deep Learning learns.

Comparison Table

Feature	Machine Learning	Deep Learning
Based On	Algorithms	Neural Networks

Data Requirement	Less Data	Huge Data
Training Time	Faster	Slower
Computing Power	Lower	Higher
Human Feature Selection	Needed	Mostly Automatic
Best For	Structured Data	Images, Audio, Video, Text
Example	Spam Detection	Face Recognition

One Diagram for Your Notes

Machine Learning:

Data
↓
Algorithm
↓
Prediction

Deep Learning:

Data
↓
Neural Network-> prediction

6. Neural Networks:

->A Neural Network is a computational model inspired by the structure and working of the human brain. It consists of interconnected artificial neurons that work together to learn patterns from data and make predictions.

->Neural Networks are the fundamental building blocks of Deep Learning. In fact, Deep Learning works by using neural networks with multiple hidden layers, which is why it is called "deep" learning.

Why Were Neural Networks Created?

->Traditional Machine Learning algorithms work well for simple and structured data. However, they struggle when dealing with large and complex data such as:

- Images
- Videos
- Speech
- Natural Language

->For many Machine Learning algorithms, humans must manually identify important features from the data.

->For example, to identify a dog in an image, a programmer may need to specify features such as:

- Ears
- Tail
- Fur

This becomes difficult for complex problems.

->To overcome this limitation, Neural Networks were developed. Instead of manually defining features, Neural Networks automatically learn important patterns directly from data.

Structure of a Neural Network:

->A Neural Network generally consists of three types of layers:

1. Input Layer:

Receives the data.

Example: Dog Image

2. Hidden Layer(s):

Processes information and learn patterns.

Examples of patterns: Edges, Eyes, Ears, Tail, Face, Shape

->A network may contain multiple hidden layers.

3. Output Layer:

Produces the final prediction.

Example:DOG

Diagram

Input Layer
↓
Hidden Layer
↓
Hidden Layer
↓
Output Layer

Why is it Called Deep Learning?

A simple Neural Network contains only a few layers.

A Deep Neural Network contains many hidden layers.

Input Layer
↓
Hidden Layer
↓
Hidden Layer
↓
Hidden Layer
↓
Output Layer

The presence of multiple hidden layers makes the network "deep," which gives rise to the term **Deep Learning**.

Types of Neural Networks:

Different neural networks are designed for different tasks:

Type	Simple Definition	Example
FNN	Basic neural network	Student result prediction
CNN	Neural network for images	Face Recognition
RNN	Neural network for sequences	Text prediction
LSTM	RNN with better memory	Language translation
Transformer	Modern network for language understanding	ChatGPT

7.TRAINING VS INFERENCE:

Training is the process in which an AI model learns patterns from data.

For example -> Suppose we want AI to identify cats.

Training Data:

Cat Image → Cat

Cat Image → Cat

Cat Image → Cat

Dog Image → Dog

Dog Image → Dog

->Thousands or millions of examples are shown.

->During training, the neural network learns:

Cats have: Whiskers, Ears, Eyes
Dogs have: Different features

Human Example:

Study
↓
Practice
↓
Learning

Training is like studying before an exam.

INFERENCE:

Inference is the process of using a **trained model to make predictions on new data.**

For example, After training:

New Image:

📷 Cat

AI predicts:

CAT

->This prediction process is called inference.

Human Example

Study
↓
Exam
↓
Answer Questions

->Inference is like writing the exam after studying.

Diagram

<u>Training Phase</u>

Training Data
↓
Neural Network Learns
↓
Trained Model

<u>Inference Phase</u>

New Data
↓
Trained Model
↓
Prediction

Real Example: ChatGPT

Training

ChatGPT learns from:

- Books
- Websites
- Articles
- Research papers

Millions of documents.

Inference:

You ask:

What is AI?

ChatGPT uses what it learned and generates an answer.

That is inference.

8. AI LEARNING METHODS:

1. Supervised Learning:

->Supervised Learning is a learning method in which AI is trained using data that already contains the correct answers (labels). The AI learns the relationship between inputs and outputs and then predicts answers for new data.

Example:

Cat Image → Cat
Dog Image → Dog

->Since the correct answers are already given, AI learns to identify cats and dogs.

Real-life Example: Spam Email Detection

2. Unsupervised Learning:

->Unsupervised Learning is a learning method in which AI is given data without answers, and it automatically finds patterns, similarities, and groups within the data.

Example:

Customer Data
↓
AI
↓
Groups customers with similar interests

->AI may automatically create groups such as students, working professionals, and senior citizens.

Real-life Example: Customer Segmentation

3. Reinforcement Learning:

->Reinforcement Learning is a learning method in which AI learns through trial and error by receiving rewards for good actions and penalties for bad actions. Over time, it learns the best strategy.

Example:

Chess Move
↓
Good Move → Reward
Bad Move → Penalty

->After many games, the AI becomes a strong chess player.

Real-life Example: Self-Driving Cars, Chess AI

4. Self-Supervised Learning:

->Self-Supervised Learning is a learning method in which AI creates its own learning tasks from unlabeled data and learns by predicting missing or next information.

<u>Example:</u>

Original Sentence:

The sky is blue.

AI creates:

The sky is _____.

Then predicts:

blue

By doing this billions of times, AI learns language, grammar, and patterns.

Real-life Example: ChatGPT, Gemini, Claude

8. AI TYPES BY INTERACTION STYLE:

1. Conversational AI:

Conversational AI is a type of AI that interacts with humans through text or voice and provides responses to questions or commands in a natural way.

User: What is Artificial Intelligence?

ChatGPT: Artificial Intelligence is a branch of computer science...

Examples: ChatGPT, Gemini, Siri, Alexa

2. Generative AI:

Generative AI is a type of AI that creates new content such as text, images, videos, music, or code based on user prompts.

Prompt:
Create an image of a tiger in space.

Output:
A new AI-generated image of a tiger in space.

or

Prompt:
Write a Python calculator program.

Output:
A new Python program.

Examples: ChatGPT, DALL-E, Midjourney, Sora

3. Agentic AI:

Agentic AI is a type of AI that can plan and perform multiple tasks on its own to achieve a goal without requiring instructions for every step.

Goal:
Plan a trip to Chennai

Agentic AI:
Search flights
↓
Compare prices
↓
Book hotel
↓
Create itinerary

Example: AI personal assistants and task automation systems.

4. Autonomous AI:

Autonomous AI is a type of AI that can observe its environment, make decisions, and perform actions independently with little or no human intervention.

Example:

Self-Driving Car

Detect road
↓
Detect vehicles
↓
Apply brakes

↓
Turn steering-> reach destination

Examples: Self-driving cars, autonomous drones, warehouse robots.